

Fundamental Principles of Thermodynamics

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February 5, 2026

1 Introduction

Classical thermodynamics provides a macroscopic description of physical systems based on a small number of experimentally measurable variables and a set of general laws. Despite its empirical origin, thermodynamics possesses a strong mathematical structure that allows the derivation of non-trivial relationships between physical quantities, independent of microscopic details.

These structural properties form the foundation for the thermodynamic potentials and the Maxwell relations developed later in this work.

2 Thermodynamic Systems and State Variables

A thermodynamic system is defined as a macroscopic portion of the universe selected for analysis. The macroscopic state of a system in equilibrium is fully specified by a set of state variables, such as pressure P , volume V , temperature T , entropy S , and internal energy U .

State variables depend only on the current equilibrium state of the system and not on the path followed to reach that state. Consequently, they can be expressed as functions of other state variables. For a simple compressible system, the internal energy can be written as

$$U = U(S, V).$$

3 Thermodynamic Equilibrium

Thermodynamic analysis is restricted to systems in equilibrium or undergoing quasi-static processes, where the system passes through a continuous sequence of equilibrium states. Only under these conditions are state variables well-defined at every stage of the process.

4 First Law of Thermodynamics

The first law of thermodynamics expresses the conservation of energy. For a closed system, the differential change in internal energy is given by

$$dU = \delta Q - \delta W,$$

where δQ is the heat added to the system and δW is the work done by the system.

For a simple compressible system performing only pressure–volume work,

$$\delta W = P dV,$$

and the first law becomes

$$dU = \delta Q - P dV.$$

Internal energy U is a state function, whereas heat and work are path-dependent quantities and are therefore not exact differentials.

5 Second Law of Thermodynamics

The second law introduces the concept of entropy and establishes the directionality of thermodynamic processes. For a reversible process, the change in entropy is defined as

$$dS = \frac{\delta Q_{\text{rev}}}{T},$$

where T is the absolute temperature.

For irreversible processes, the entropy change satisfies the inequality

$$dS > \frac{\delta Q}{T}.$$

Entropy is a state function and plays a central role in equilibrium thermodynamics and in the definition of thermodynamic potentials.

6 Exact and Inexact Differentials

A differential quantity dX is said to be exact if it corresponds to the differential of a state function X . Heat δQ and work δW are inexact differentials because they depend on the path taken during a process.

For a state function $f(x, y)$ with continuous second derivatives, the equality of mixed partial derivatives holds:

$$\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 f}{\partial y \partial x}.$$

This mathematical property is fundamental to thermodynamics and forms the basis for the derivation of the Maxwell relations.

7 Simple Compressible Systems

In this work, attention is restricted to simple compressible systems, characterized by the variables (P, V, T, S) and described by equations of state of the form

$$P = P(T, V).$$

This restriction allows for a clear and systematic derivation of thermodynamic identities while maintaining physical relevance and computational tractability.